Solvency Field Theory V13.1-V3 Numerical-G Companion v1.1

Conditional Zero-Continuous-Parameter Prediction from the Electron Closure Operator

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2026-07-26 - provenance-audited revision

DOCUMENT STATUS
Selected-receipt result companion v1.1. This revision seals the threshold-recurrence provenance, corrects the published beta edge at the fifteenth decimal place, adds CODATA central-value drift context, and updates the reproduction firewall. It attaches to the revised SFT V13.1-V3 Gravity Derivation Canon and does not replace the structural field-equation source of truth.

RESULT AT A GLANCE

Conditional SFT prediction

$$G_0 = 6.674306513706680 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$$

0.976 ppm above the 2022 CODATA central value; current CODATA relative standard uncertainty is 22.5 ppm.

Claim boundary: conditional on the explicit minimal optical-normalization postulate c = 1. The full first-principles derivation remains open by one universal dimensionless representation normalization.

MASTER VERDICT
PASS - CONDITIONAL ZERO-CONTINUOUS-PARAMETER NUMERICAL-G PREDICTION FROZEN; Q*/BETA PROVENANCE SEALED; FULL FIRST-PRINCIPLES DERIVATION OPEN BY ONE UNIVERSAL OPTICAL NORMALIZATION.

Companion sentence: *One residual, one representation choice, no hidden retuning.*

Contents

Executive result statement

The revised SFT V13.1-V3 gravity canon structurally closed the selected local Einstein-form branch and derived the coupling location G = a_phi c^3 / hbar, while proving that the then-current scale-free equations did not select the absolute root area. This companion addresses that specific normalization debt through the later electron closure/operator program. It does not rerun the V13.1-V3 field equation, hostile gravity tests, strong-field program, or cosmology program.

The new result is best stated in two layers. Internally, SFT conditionally predicts the electron's dimensionless gravitational share epsilon_e from the six-sector register, the limiting electron recurrence, and the rank-three settlement availability operator. Externally, the independently measured inertial electron mass provides the SI anchor, so numerical G follows without using measured G during candidate construction.

$$\Gamma_e^{(0)} = \left(1 - \frac{q_*}{3}\right)\left(\frac{e^{3L_{\rm eff}}}{2\pi}\right)^2, \qquad G_0 = \left(\varepsilon_e^{(0)}\right)^2 \frac{\hbar c}{m_e^2}$$

With L_eff = 9.097814727066048 and the independently selected stable recurrence q_* = 0.091710909655065, the identity optical representation c = 1 gives Gamma_e = 1.250089141355940e+22, epsilon_e = 4.185464261523565e-23, and G_0 = 6.674306513706680e-11 m^3 kg^-1 s^-2.

The result is 0.976 parts per million above the current CODATA central value. That numerical closeness must not be overstated: the current relative standard uncertainty of measured G is about 22.5 ppm, so the present experiment is consistent with the prediction but does not verify it at one-part-per-million precision.

The final claim boundary remains explicit. The electron-specific quotient is derived conditionally from frozen operator categories. The universal identity map from baseline six-register optical capacity to the coefficient Gamma is adopted as the minimal representation postulate c = 1; current frozen doctrine still permits Gamma = c chi_e B_6 for any positive universal c. Therefore this is a serious, falsifiable, zero-continuous-parameter conditional prediction, not a completed first-principles derivation of the SI number G.

Document control and relationship to V13.1-V3

This document is a result companion to SFT V13.1-V3, not a replacement canon. The V13.1-V3 source-of-truth document remains governing for the settlement carrier, boundary-complete source current, ordered-history composition, address naturality, local state closure, Einstein-form selected branch, hostile no-retuning record, and structural relation G = a_phi c^3 / hbar.

The companion changes one later truth status only: V13.1-V3 correctly said that its own scale-free structural stack did not derive numerical G. The later electron program now supplies a conditional dimensionless electron-to-root ratio. When that comparison-separated dimensionless result is combined with an independently measured non-gravitational electron mass, a numerical G prediction becomes possible. The strict open debt moves from “no numerical candidate” to “one universal optical representation normalization remains unproved.”

Three branches must now remain distinct:

- Fully blind symbolic branch: no absolute mass and no measured G; a_phi and G remain symbolic.

- Electron-anchored prediction branch: derive epsilon_e without measured G, then use the independently measured inertial electron mass to predict G.
- G-correspondence control: use measured G once to calibrate c; this is a calibration and may never be called a prediction of G.

The source-of-truth discipline remains unchanged: failed mechanisms are preserved, corrected definitions supersede earlier interpretations, and no sector-specific retuning is allowed.

Part I - What is being claimed

1. Primary claim

Under the explicit minimal optical-normalization postulate c = 1, the frozen SFT electron closure/operator stack predicts the dimensionless electron gravitational share and therefore predicts numerical G from the independently measured inertial electron mass.

$$G_0 = 6.674306513706680 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$$

The predicted central value is 0.976 ppm above the 2022 CODATA central value and lies within the current 22 ppm relative standard uncertainty of measured G.

2. Claim hierarchy

Status class	Content
Derived / frozen	L_eff; the Z3 generation operator; the six-sector determinant D6; the limiting electron recurrence q_*; amplitude-level withholding; normalized-trace availability chi_e.
Derived conditionally	The electron coefficient Gamma_e once the direct optical identity representation is selected.
Explicit minimal postulate	The universal optical representation is identity: c = 1.
Conditional prediction	epsilon_e and numerical G0 using the independently measured inertial electron mass.
Calibration control	c_anchor = 0.999999512031011 if measured G is used once; never a prediction of G.
Open theorem	An independent reason that the primitive optical representation must have c = 1.

3. What is not claimed

- SFT has not derived an SI metre or kilogram from pure dimensionless numbers.
- The value c = 1 has not been proven uniquely by the current frozen operator stack.
- Agreement with the present central value of G is not a one-ppm experimental confirmation because measured G is uncertain at roughly 22 ppm.
- The electron result does not yet derive the muon, tau, neutrino, quark, hadron, W, Z, or Higgs-like sectors.

- The result does not complete strong-field gravity, the full post-Newtonian hierarchy, black-hole microphysics, dense matter, or cosmological write dynamics.

Part II - Minimal derivation spine

4. Frozen operator scale

The generation transfer operator is the covariant Z3 Laplacian

$$U_\beta = I - \tfrac{1}{2}(e^{i\beta}S + e^{-i\beta}S^\dagger), \qquad u_k(\beta) = 1 - \cos\left(\beta + \tfrac{2\pi k}{3}\right)$$

and the maintained rigidity operator uses the previously frozen dynamic scale

$$L_{\mathrm{eff}} = 3\pi - \tfrac{1}{2}\ln\left(\tfrac{25}{13}\right) = 9.097814727066048, \qquad K_\beta = \tfrac{3\pi}{4}I + \tfrac{L_{\mathrm{eff}}}{4}U_\beta$$

The complete family law acts on a six-sector register. Its amplitude-transfer pseudo-determinant is

$$D_6 = e^{3L_{\mathrm{eff}}} = 7.134970798893849 \times 10^{11}$$

The minimal identity optical baseline is the squared circumference-normalized capacity

$$B_6 = \left(\tfrac{D_6}{2\pi}\right)^2 = 1.289509848425358 \times 10^{22}$$

G13L38E later repaired the area dictionary: B6 is not automatically raw write area or support-radius squared. The final result uses B6 as baseline optical capacity under the explicit identity representation c = 1.

5. Limiting electron socket and stable recurrence

The charged electron is the n = 3 weak-lane maintenance channel at the exact all-twelve matter/light boundary. The limiting socket is selected independently of the electron mass by

$$P_{\mathrm{coh}}(q_*, \pi)\,u_* = \tfrac{1}{3}, \qquad P_{\mathrm{coh}}(q, \pi) = \left(\tfrac{1-q}{1+q}\right)^2$$

with the recurrence

$$q = \exp\left[-K_* P_{\mathrm{coh}}(q, \pi)\right], \qquad K_* = \tfrac{3\pi}{4} + \tfrac{L_{\mathrm{eff}}}{4}u_*$$

The beta edge is not an independent input to the production calculation. At the exact matter/light boundary, P_coh(q, pi) u = 1/3 eliminates u from the recurrence and gives q = exp[-(3 pi/4) P_coh(q, pi) -

L_eff/12]. The unique stable root is q_* = 0.091710909655065... . Only afterward are u_* and the conjugate phase edges beta = pi/3 +/- 0.021432753081800... reconstructed from the weak-lane geometry.

$$q_* = 0.091710909655065$$

has recurrence-map derivative 0.883865 and is stable. The second root q = 0.137772... has derivative greater than one and is rejected without reference to the electron mass or measured G.

5A. Threshold-recurrence provenance receipt

The successful quotient did not supply its own threshold. The recurrence, rigidity floor, and matter/light condition were frozen in earlier gates. The exact dependency is recurrence + threshold -> q_* -> u_* -> beta; neither the measured value of G nor the electron gravitational correspondence enters those steps.

The project was not historically ignorant of measured G. The defensible claim is comparison separation: the final candidate formula was generated from frozen internal equations before its numerical comparison. Unqualified “blind derivation” language is therefore replaced by “comparison-separated” or “target-excluded calculation.”

Receipt / date	Frozen contribution	Target use	Chronology role
MWO-G11B / 2026-07-22	q = exp[-K P_coh] and P_coh(q, delta)	None	Recurrence predates final quotient
MWO-G11D / 2026-07-22	K = 3pi/4 + (L_eff/4)u_k	None	Root-plus-lane rigidity predates final quotient
UNI-G13L10E / 2026-07-23	Matter/light threshold P_coh u > 1/3	None for threshold law	Threshold predates beta edge
UNI-G13L15E / 2026-07-24	All-twelve edge |beta-pi/3| < 0.021432753082	Holdout only in later diagnostic	Edge predates residual formula
UNI-G13L25E / 2026-07-24	Exact edge 0.021432753081800; 0.999-edge coincidence rejected	Holdout used to falsify maps	Correction predates G13L39E
UNI-G13L39E / 2026-07-26	chi_e = 1 - q_*/3 and final comparison	Comparison only after formula freeze	First successful occupied-state quotient

6. Occupied-state availability

The universal settlement register has rank three, while one physical electron occupies a rank-one maintenance state. The retained monodromy quantity q_* is an amplitude-level object. Because same-write support is a linear pre-norm projection, the retained occupied component is withheld before any Born/intensity readout:

$$A_e = I_3 - q_* P_e, \qquad \mathrm{rank} P_e = 1$$

Basis invariance, linearity over orthogonal settlement directions, continuity, and normalization on I3 uniquely select normalized trace as the scalar handoff:

$$\chi_e = \frac{1}{3}\mathrm{Tr} A_e = 1 - \frac{q_*}{3} = 0.969429696781645$$

This closes the earlier 3.15% baseline excess without inserting a fitted profile factor. The withheld share is the retained amplitude in one occupied direction averaged over the three settlement directions.

7. Optical coefficient and support geometry

Under the minimal identity optical representation, the electron coefficient is

$$\Gamma_e^{(0)} = \chi_e B_6 = 1.250089141355940 \times 10^{22}$$

The corrected native dictionary is

$$\mathcal{A} = \frac{A_{\text{write}}}{a_\phi}, \qquad X = \frac{R^2_{\text{write}}}{a_\phi}, \qquad \Gamma = \lambda \mathcal{A}, \qquad X = \frac{\pi}{4}\Gamma$$

The frozen lower optical branch relates Gamma to the route/support ratio rho through the radial optical saddle a(rho):

$$\Gamma(\rho) = \frac{64\alpha_\theta \rho^2}{\pi a(\rho)}, \qquad \alpha_\theta = \frac{1}{16\pi^2}$$

Solving this monotone branch gives rho_e approximately 1.205621e+11. The dimensionless electron gravitational share is then

$$\varepsilon_e^{(0)} = \frac{\sqrt{a_\phi}}{2R_e} = \frac{1}{\rho_e \sqrt{\pi \Gamma_e^{(0)}}} = 4.185464261523565 \times 10^{-23}$$

8. SI handoff

The experimentally measured quantity used here is the inertial electron mass, not gravitational attraction. With the mass-equivalence and species-universality bridge from the cross-settlement program,

$$G_0 = (\varepsilon_e^{(0)})^2 \frac{\hbar c}{m_e^2}$$

Using exact h and c and the 2022 CODATA electron mass m_e = 9.1093837139 x 10^-31 kg yields the result at the top of this paper. Measured G was not used to construct L_eff, D6, q_*, chi_e, Gamma_e, rho_e, or epsilon_e.

Part III - Numerical result, provenance, and experimental comparison

9. Numerical receipt

Quantity	Value	Role
L_eff	9.097814727066048	Frozen dynamic generation scale
D6	7.134970798893849e+11	Six-sector amplitude-transfer determinant
B6	1.289509848425358e+22	Minimal identity optical baseline

Quantity	Value	Role
q_*	0.091710909655065	Unique stable limiting electron recurrence
chi_e	0.969429696781645	Normalized settlement availability
Gamma_e^(0)	1.250089141355940e+22	Conditional electron optical coefficient
rho_e	1.205621361455173e+11	Large-route optical solution
epsilon_e^(0)	4.185464261523565e-23	Dimensionless electron gravitational share
G0	6.674306513706680e-11	m^3 kg^-1 s^-2

10. Comparison with current constants

Comparison item	Value	Interpretation
SFT conditional prediction	6.674306513706680e-11	Identity optical representation c = 1
2022 CODATA G	6.67430e-11	Standard uncertainty 1.50e-15
Central-value difference	+6.513707622749502e-17	+0.976 ppm
CODATA relative uncertainty	2.247427e-05	22.5 ppm

The prediction is consistent with the present measurement. It is not experimentally distinguished from nearby values inside the current G uncertainty. A future improvement in G measurement below the ppm level would directly test the fixed identity prediction.

10A. CODATA central-value drift

The sub-ppm central-value offset must be read against the historical movement and uncertainty of recommended G. Official NIST/CODATA values moved by 68.93 ppm from 2010 to 2018. The 2018 and 2022 central values are identical. The conditional SFT value lies within the one-standard-uncertainty interval of every release shown below.

CODATA	Central G (SI)	Std. uncertainty	Relative u (ppm)	SFT G0 - central (ppm)
2010	6.67384e-11	0.00080e-11	119.87	+69.90
2014	6.67408e-11	0.00031e-11	46.45	+33.94
2018	6.67430e-11	0.00015e-11	22.47	+0.976
2022	6.67430e-11	0.00015e-11	22.47	+0.976

Accordingly, this paper claims consistency with the present CODATA interval and a 0.976-ppm central-value offset. It does not claim experimental confirmation at sub-ppm precision.

11. Conditional root-scale outputs

For reference, the same conditional result corresponds to the universal write-area and root quantities

Quantity	Conditional value	Meaning
a_phi^(0)	2.612282853400299e-70 m^2	Universal write-area normalization under c = 1
sqrt(a_phi^(0))	1.616255813106421e-35 m	Root length

Quantity	Conditional value	Meaning
m_phi^(0)	2.176433280685431e-08 kg	Root mass from sqrt(hbar c / G0)

These are consequences of the conditional G prediction, not separate independent predictions or additional fits.

Part IV - Comparison separation, corrections, and no-go record

12. Candidate-generation firewall

The recurrence and threshold ingredients predate the successful occupied-state quotient. In the production script, q_* is solved from the reduced threshold equation and beta is derived afterward; no beta decimal, measured G, or electron gravitational target enters candidate generation. The final comparison is separated in a dedicated module. This is comparison separation, not a claim that the research team lacked historical knowledge of the target.

The identity branch is retained even though an anchor-adjusted c can match the central value exactly. The c = 1 value and its discrepancy may never be overwritten by the calibration control.

13. Alternative scalar controls

Candidate	Definition	Gamma rel. error	G rel. error
Amplitude normalized trace	1 - q/3	-4.880e-07	+9.759e-07
Amplitude quadratic mean	Schatten-2 mean	+9.934e-04	-1.984e-03
Amplitude geometric mean	det^(1/3)	-1.017e-03	+2.037e-03
Identity / no withholding	1	+3.153e-02	-6.021e-02
Intensity normalized trace	1 - q^2/3	+2.864e-02	-5.491e-02

Numerical uniqueness is not used as the proof. The normalized trace is selected by the linear operator axioms; the control table shows that the independently selected scalar is also the only preregistered standard candidate closing at the sub-ppm level.

14. Correction firewall

The result must be read through the corrected G13L38E area dictionary. Earlier gates sometimes used Gamma for raw write area, support-radius squared, and optical coefficient interchangeably. Those interpretations are superseded. The live result uses B6 as baseline optical capacity, then applies the occupied-state quotient and the explicit identity map into Gamma.

The earlier boundary-metric residual mu = 0.984596... is withdrawn as an artifact of that convention collision. Under the corrected boundary-radius branch the required mu would be 0.872575..., but that branch is not the final result.

15. Why lambda was not a hidden profile integral

The G12U normalized footprint was exhaustively integrated. Its shape is identical for every positive lambda. Exact profile quantities such as C_G = 3/4 and C_2 = pi/8 determine shape functionals after rescaling, but they do not identify which effective area definition is the native A_write. The profile therefore carries zero information about lambda as a constitutive conversion.

This closes the cheapest alternative explanation: the final quotient is not a forgotten shape integral.

16. Independent normalization no-go classes

No-go class	Lasting result
Homogeneous stability	Scale variations preserve the dimensionless equations; no finite scale minimum.
Formation overlap and one-write action	The radius cancels or only dimensionless ratios are selected.
Far-field pair gravity	The monopole sees conserved mass and is blind to internal support calibration.
Near-field contact and slot spacing	Resolved geometry changes forces, but integer slot gaps rescale with the same ruler.
Profile reconstruction	The normalized profile has zero Fisher information about lambda.
Representation family	Gamma = c chi_e B6 preserves current structural axioms for every positive universal c.

Together these results justify stopping further homogeneous scale, profile, formation, and contact searches. The numerical-G program reopens only if a genuinely independent operator or representation theorem fixes c.

Part V - Selected receipt chain

17. Inclusion rule

This companion intentionally does not reproduce the full electron/positron archive. A run is included only if it satisfies at least one of three conditions: it supplies a live equation or numerical input used in the final result; it repairs a definition required to state the result correctly; or it proves a no-go that fixes the final claim boundary.

Exploratory coefficient scans, superseded area conventions, failed formation models, and intermediate routes are preserved in the full archive but omitted here. Omitting them from this result paper does not erase them; the correction and no-retuning ledgers remain binding.

18. Live derivation receipts

Receipt	Release	Role in this paper
V3-CAN	SFT V13.1-V3 Gravity Derivation Canon	Structural field equation, G = a_phi c^3/hbar, normalization debt and claim firewall.
G13L20E	Generation operator and maintenance monodromy	L_eff, U_beta, K_beta, six-family maintenance architecture, electron socket.
G13L21E	Canonical log-monodromy action	q as retained amplitude and additive history action C_hist = -ln q.
G13L27E	Geodesic-bundle optical saddle	Finite bundle optical coefficients and support completion.

Receipt	Release	Role in this paper
G13L28E	Framing optical stress	Native Gamma(rho) route/support relation.
G13L31E	Primitive-area uniqueness and route map	Monotone Gamma-to-rho solution and electron epsilon correspondence firewall.
G13L32E	Six-sector support determinant	D6 = exp(3 L_eff) and B6 candidate.
G13L33E	State occupancy versus law register	Rank-one occupied state distinct from rank-six universal law.
G13L38E	Area-convention repair	Separates raw area, radius squared, and optical Gamma.
G13L38F	Profile-integral exhaustion	Proves lambda is not recoverable from the normalized profile alone.
G13L39E	Occupied-state/register handoff	q_*, chi_e = 1 - q_*/3, blind sub-ppm candidate and controls.
G13L40E	Retained-amplitude theorem	Closes amplitude versus intensity; proves one universal c remains.
G13L41E	G-program freeze	Freezes identity prediction, one-anchor firewall, and particle-atlas open.
AUD-G13L42G	Threshold-recurrence provenance and CODATA-drift audit	Seals q*/beta dependency, chronology, decimal correction, and comparison language.

19. Supporting no-go receipts retained in the slim pack

The selected receipts pack also retains the adjudication ledgers from G13L35E-G13L37E because they establish, respectively, the phase-to-length modulus, far-field interior blindness, and near-field slot/contact cancellation. Their full exploratory scans are not reproduced in the main paper.

Part VI - Public claim registry

20. Claims this companion may make

- SFT V13.1-V3 derives the structural location of gravitational coupling and the selected Einstein-form gravity branch.
- The later electron operator program derives a specific dimensionless electron gravitational share conditional on the minimal identity optical representation.
- Under c = 1, SFT makes a zero-continuous-parameter conditional prediction G0 = 6.674306513706680 x 10^-11 m^3 kg^-1 s^-2.
- The predicted central value is consistent with current measured G and differs from the 2022 CODATA central value by 0.976 ppm.
- The electron-specific residual is explained by retained amplitude in one occupied direction averaged over the three settlement directions.
- The one-anchor control is separate and cannot overwrite the identity prediction.

21. Claims this companion may not make

- SFT has unconditionally derived numerical G from pure dimensionless axioms.
- The identity optical normalization c = 1 has been proven uniquely.
- The result has been experimentally confirmed at one-part-per-million precision.
- Measured G may be used to set c and then presented as an independent prediction.
- The electron result already derives the full particle spectrum.
- The full strong-field, black-hole, dense-matter, or cosmological programs are complete.

Part VII - Open normalization theorem and future particle program

22. Remaining gravity open

The only live normalization debt is an independent theorem fixing the primitive optical representation:

$$\Gamma_{\text{base}} = cB_6 \quad \Longrightarrow \quad c = 1 \text{ independent operator theorem}$$

Until such a theorem exists, c = 1 remains the explicit minimal postulate and c_anchor remains a calibration control. Further homogeneous scale, profile, formation, or contact gates are stopped.

23. Geometry-to-particle atlas

The next program is not another G search. It is the geometry-to-particle atlas: derive every SFT-locally-real particle class from admissible closure geometry rather than importing the particle list.

The immediate make-or-break gate is the charged-lepton trilogy:

$$\text{same charged– lepton geometry} + \text{different generation lane} \longrightarrow m_e : m_\mu : m_\tau$$

The universal optical normalization c cancels from common-branch mass ratios. Therefore the muon/electron and tau/electron predictions can be tested without reopening numerical G.

Priority	Workstream	Primary target
1	Charged leptons	Blind m_mu/m_e and m_tau/m_e from lane-specific recurrence and availability.
2	Neutral leptons	Neutral handed closures, mass ordering, dimensionless splittings, and mixing map.
3	Colored irreducible closures	Color ports, fractional winding, incomplete-address rejection, confinement.
4	Hadronic composites	Proton/neutron geometry, overlap energy, spin, charge, and stability.
5	Branch-changing/rigidity resonances	W, Z, and Higgs-like eigenmodes and widths.
6	Completeness theorem	Finite survivor list and exclusion of nearby unobserved free geometries.

Final verdict

The clean source-of-truth statement is:

> **Starting from the structurally closed SFT V13.1-V3 gravity branch and the later locally real electron closure/operator stack, SFT derives the six-sector baseline, the limiting electron recurrence, the rank-one occupied-state availability quotient, and the dimensionless electron gravitational share under the explicit minimal optical**

identity representation. With the independently measured inertial electron mass, this yields the conditional prediction G0 = 6.674306513706680 x 10^-11 m^3 kg^-1 s^-2, 0.976 ppm above the current CODATA central value and inside its present uncertainty. The result is zero-continuous-parameter and falsifiable, but not a completed first-principles derivation because one universal representation normalization c = 1 remains a postulate rather than a theorem. The G program is frozen at that boundary, and the active program moves to the geometry-to-particle atlas.

Final status: CONDITIONAL NUMERICAL-G PREDICTION FROZEN; FULL DERIVATION OPEN BY ONE UNIVERSAL OPTICAL NORMALIZATION; PARTICLE-GEOMETRY ATLAS READY.

Appendix A - Compact formula ledger

Item	Formula
Generation eigenvalues	u_k(beta) = 1 - cos(beta + 2 pi k/3)
Rigidity operator	K_beta = 3 pi/4 I + (L_eff/4) U_beta
Six-sector determinant	D6 = exp(3 L_eff)
Optical baseline	B6 = (D6/2 pi)^2
Coherence	P_coh(q, pi) = ((1-q)/(1+q))^2
Electron threshold	P_coh(q_*, pi) u_* = 1/3
Recurrence	q_* = exp[-K_* P_coh(q_*, pi)]
Availability	chi_e = (1/3) Tr(I3 - q_* P_e) = 1 - q_*/3
Conditional optical coefficient	Gamma_e^(0) = chi_e B6
Optical route map	Gamma(rho) = 64 alpha_theta rho^2 / [pi a(rho)]
Electron share	epsilon_e = 1/[rho_e sqrt(pi Gamma_e)]
Numerical G	G0 = epsilon_e^2 hbar c / m_e^2

Appendix B - Selected artifact ledger

The selected-receipts ZIP preserves the complete artifact filenames and copied source files. This compact ledger records the receipt label and immutable SHA-256 only.

Receipt	SHA-256
V3-CAN	c0922c79e5c67ebf67cf55d59d0ff6de41e36d961e41010f47183742466707d4
G13L20E	9698c548be576ef58a7d1c6ea089f329b9028770fb5c54acda016524fad46011
G13L21E	7d6c7f35d63e46082d7ad2927582e6527bdca9166839869f6541d6f4a68e32de
G13L27E	573b8b05f50c3e98a48981d8677010130c64b4ab80b876540bd8edde2263b00b
G13L28E	ab251a7fae9d2adfde449e2e3051e8474b7c6e536587209158996f5a75d79490
G13L31E	476225299b9dfcb63182691857006d774283076aae8557754ac94076aa773591
G13L32E	103047e212e56a5ed212d7b1efa4d518cbc1e45710582048dbdbc6b43d302b24
G13L33E	b2eee6899475eebc2963c915038daf9e584417c7658f67eb3f82a2ba85e4f6ff
G13L35E	d1485eb24389a85352eeec7016f7294ac3fba576e10535a562999b61b089deb0
G13L36E	d3c62e637175ba217c6cb1925b98dd4a84f73d479dc1d60262615b17e0687782
G13L37E	32edf36bd76c4ec8de3af6bb2f69ba0b7912c6e56a4bb1c949d0cbd2e210f0d1
G13L38E	5c5eba614eb7f2b5e4d6432db2c44bd16876c88b8d6e5fcfbdf72259444dbaa1
G13L38F	24dc0a0c840b56ced28fc5367140c1f5d8e9be3c5b8c9f81ac0841efda915a27
G13L39E	a989553b88b1cf218e865feffd83785f1c7f1838be0b318df39e809eddd8f026

Receipt	SHA-256
G13L40E	96a661dab39314b62b2540ca612afb515c67b8b2e661422c4ba7ec4b286dcca8
G13L41E	9fc6fc500f39c326b61933aa0087fda610b33ee8b411b0a045aa8670aab13000
AUD-G13L42G	251d38fae4f2241349f2aaafb2c2e97d5f7cd3833333384da8b5ada11d0bbd77

Appendix C - External comparison references

[NIST-G-2022] NIST/CODATA, 2022 complete listing: physics.nist.gov/cuu/Constants/Table/allascii.txt

[NIST-G-2010] NIST/CODATA 2010 archive: physics.nist.gov/cuu/Constants/ArchiveASCII/allascii_2010.txt

[NIST-G-2014] NIST/CODATA 2014 archive: physics.nist.gov/cuu/Constants/ArchiveASCII/allascii_2014.txt

[NIST-G-2018] NIST/CODATA 2018 archive: physics.nist.gov/cuu/Constants/ArchiveASCII/allascii_2018.txt

[NIST-me] NIST/CODATA, “electron mass,” 2022 CODATA recommended value. physics.nist.gov/cgi-bin/cuu/Value?me=

[V3-CAN] SFT V13.1-V3 Gravity Derivation Canon, revised source-of-truth release, 2026-07-22.

The external constants are used only for the final SI comparison. All native SFT candidate selection is governed by the internal receipt chain and its comparison/anchor firewalls.